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Determination of Porosity and Hydrocarbon Saturation for Heavy Oil Reservoirs Using Dielectric and NMR Techniques

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Abstract

Oil and gas exploration requires precise evaluation of various parameters, among which porosity and saturation are the most crucial and fundamental properties. Conventional logging methods like neutron, density, and sonic logs used for porosity measurements experience various challenges in heavy oil reservoirs. Thus, this study aims to reduce these challenges and uncertainty in estimating porosity and saturations in heavy oil reservoirs by utilizing specialized techniques including nuclear magnetic resonance (NMR) and multifrequency dielectric dispersion approaches.

The study uses carbonate and sandstone rocks having an average permeability of 200 mD and porosity of 15% and 18.5%, respectively. The samples were then saturated with 3wt.% KCl brine followed by NMR, gravimetric, and dielectric measurements for porosity determination. Next, the samples were saturated with heavy oil that has a viscosity of more than 4500 cP using high-pressure and high-temperature (HPHT) flooding experiments. The NMR and dielectric measurements were again taken at different oil saturations, followed by CT scan analysis to visualize the oil and brine distribution inside the samples. Finally, the porosity and hydrocarbon saturations were compared with all the determination techniques.

The result shows a significant impact of heavy oil on rock porosity estimation and dielectric measurements. The NMR porosities at fully brine-saturated samples were 16% and 19.3% for carbonate and sandstone, respectively. After heavy oil injection up to irreducible water saturation (S_{wirr}), the total porosity was decreased to 12.5% and 15.3% for carbonate and sandstone samples, likely due to heavy oil components occupying part of the pore spaces. When the heavy oil saturation was increased by injecting more oil, the measured porosities were around 7.9% and 9.8%, respectively. The reduction in total porosity might be due to the entrapment of heavy oil components in the rock samples. This was also confirmed by NMR T_2 shift and CT scan analysis in which patches of dense components were seen after reducing the heavy oil saturation. Dielectric measurements followed a similar trend. For carbonate, the permittivity value

decreased from 10.15 at S_{wi} (100%) to 6.19 at S_{wirr} (45%) and slightly increased to 6.30 at S_o of 53% (S_w is 47%). For sandstone, the permittivity reduced from 8.27 at S_{wi} (100%) to 3.76 at S_{wirr} (25%) and further decreased to 3.64 at S_o of 34% (S_w is 66%). The conductivity profiles showed a decreasing trend with more oil saturation. These findings highlight the challenges in porosity determination in the presence of heavy oil and propose a framework for achieving more accurate porosity estimation in heavy oil reservoirs.

This study uniquely combines NMR and multifrequency dielectric techniques to quantify porosity and hydrocarbon saturation at different saturation stages in heavy-oil reservoirs. The resulting workflow improves rock-fluid characterization in both carbonate and sandstone formations, clarifying how heavy oil distributes within the pore system. These insights enable more precise steam-flood design and other EOR strategies, mitigating production challenges and improving recovery efficiency from heavy oil reservoirs.

Keywords: Carbonate and sandstone formation, Heavy oil reservoirs, Porosity estimation, NMR (Nuclear Magnetic Resonance), Multifrequency dielectric dispersion.

Introduction

In reservoir characterization, porosity and hydrocarbon saturation are crucial factors as they determine the formation's storage capacity and its potential hydrocarbon productivity (Kenyon, 1992; Clennell et al., 2018). Conventional logging techniques, such as neutron porosity and bulk density, indirectly estimate these properties but face some limitations in heavy oil reservoirs (Freedman and Heaton, 2004; Schmitt et al., 2011). Because of high hydrogen content and water-like density of heavy oil, neutron and density logs often incorrectly identify oil as water, resulting in overestimated apparent porosities and ambiguous fluid interpretations (Freedman and Heaton, 2004; Valiakhmetov et al., 2019). Additionally water saturation calculation using resistivity log is not reliable in heavy oil reservoir, particularly in zones consisting of fresh formation water and altered wettability, as standard Archie parameters (cementation exponent, m , and saturation exponent, n) may no longer be valid (Schmitt et al., 2011; Del Ángel Morales et al., 2013; Herlinger, 2019). These limitations have motivated the use of alternative petrophysical techniques that directly assess the pore fluids and rock-fluid interactions, most notably nuclear magnetic resonance (NMR) and dielectric dispersion measurements (Freedman and Vogiatzis, 1979; Kleinberg and Jackson, 2001).

NMR and dielectric tools offer complementary insights and are considered "dispersive" measurements because their responses vary with measurement frequency. Also, they are highly sensitive to pore-scale fluid interactions with the rock matrix (Kenyon, 1992; Freed et al., 2018). NMR logging has long been used to quantify porosity and pore-size distributions, as well as to identify movable fluids in situ (Kenyon, 1992; El-Husseiny & Knight, 2017; Isah et al., 2021). Unlike neutron or density logs, NMR directly measures the liquid-filled pore volume and distinguishes fluid types based on their characteristic relaxation and diffusion behavior, a critical advantage in heavy oil reservoirs where viscous oil exhibits distinctive NMR signatures (Freedman and Heaton, 2004; Rakhmatullin et al., 2020). Advanced NMR techniques can even quantify heavy-oil viscosity via diffusion-relaxation methods and provide indicators of rock wettability that conventional logs cannot (Freedman & Vogiatzis, 1979; Moro et al., 2021; Al-Garadi et al., 2022).

Dielectric dispersion logging was introduced in the 1970s to provide direct measurement of water-filled porosity and reduce uncertainties in saturation estimates, particularly under challenging conditions such as fresh or brackish water zones, clay-rich formations, and heavy oil reservoirs (Freedman and Vogiatzis, 1979; Schmitt et al., 2011; Del Ángel Morales et al., 2013). This technique leverages the strong contrast between water's high dielectric permittivity and the much lower values of hydrocarbons and rock matrices, offering an accurate alternative to resistivity-based methods for determining water saturation (Davis and Annan, 1989; S. M. Al-Ofi et al., 2018). Dielectric measurements exhibit inherent frequency dependence and are sensitive to interfacial polarization. For instance, clay-bound water strongly increases dielectric constants at low frequencies and alters dispersion behavior (Topp et al., 1980; Davood Namdar-Khojasteh, 2012;

Freed et al., 2018). Recent studies have used multifrequency dielectric logging to assess the distribution of fluids which can help in determining the wettability of rock and providing detailed insights into pore-fluid arrangements that have been difficult to analyze using NMR only (Al-Ofi et al., 2018; Posenato Garcia et al., 2022; Oshaish et al., 2023).

Leveraging the unique strengths of NMR and dielectric measurements techniques can significantly reduce the uncertainty of petrophysical interpretations (Clennell et al., 2018; Hassan et al., 2022). The combined NMR and dielectric approach has been proven successfully across numerous challenging scenarios that include characterization of clay-rich mud rocks (Clennell et al., 2006), tracking organic maturation in shale reservoirs (Hassan et al., 2022), assessing wettability alteration in carbonate formation (Venkataramanan et al., 2014), and confirming residual oil saturation in depleted heavy oil reservoirs (Valiakhmetov et al., 2019). Such studies demonstrate how a combined analysis can give more accurate and comprehensive petrophysical insights. NMR measurements give detailed information about pore-size distributions and fluid mobility while the dielectric tool directly quantifies water content and also reflects wettability conditions. This synergy assists in eliminating uncertainties that are commonly associated with the single-tool interpretations (El-Husseiny and Knight, 2017; Oshaish et al., 2023). Even under unconventional conditions, reliable saturation estimation is achievable by leveraging these frequency-dependent measurements to distinguish viscous hydrocarbons from bound water (Freedman and Heaton, 2004; Valiakhmetov et al., 2019; Al-Ofi et al., 2022). Consequently, integrated NMR and dielectric logging methods are gaining widespread use for reducing uncertainties in the evaluation of heavy oil reservoirs (Herlinger, 2019; Elsayed et al., 2022).

To address the limitations of conventional methods for estimating porosity and saturation in heavy-oil reservoirs, this study integrates laboratory nuclear magnetic resonance (NMR), multifrequency dielectric dispersion, and gravimetric analyses to quantify porosity and hydrocarbon saturation in sandstone and carbonate cores across a range of heavy-oil saturations. X-ray computed tomography (CT) is also employed to visualize pore structure and identify trapping of viscous components within the rock matrix. Together, these comprehensive techniques improve the accuracy of pore-structure and fluid-saturation evaluation, yielding more reliable reservoir characterization to support heavy-oil recovery.

Methodology

A guided experimental design process illustrated in Fig. 1 was developed to help investigate the influence of heavy oil on the porosity and hydrocarbon saturation measurements of carbonate and sandstone cores. A high permeability (approximately 200 mD) Indiana limestone and Berea sandstone core plugs were selected to mimic and represent the typical heavy-oil reservoir formation. Each core plug measured 1.5 inches in diameter and 3 inches in length. Before any fluid introduction, the cores were oven-dried at 75 °C for 24 hours to remove residual moisture. After cooling in a desiccator to avoid atmospheric moisture uptake, each core's dry mass was recorded to establish a baseline for subsequent mass-based saturation calculations.

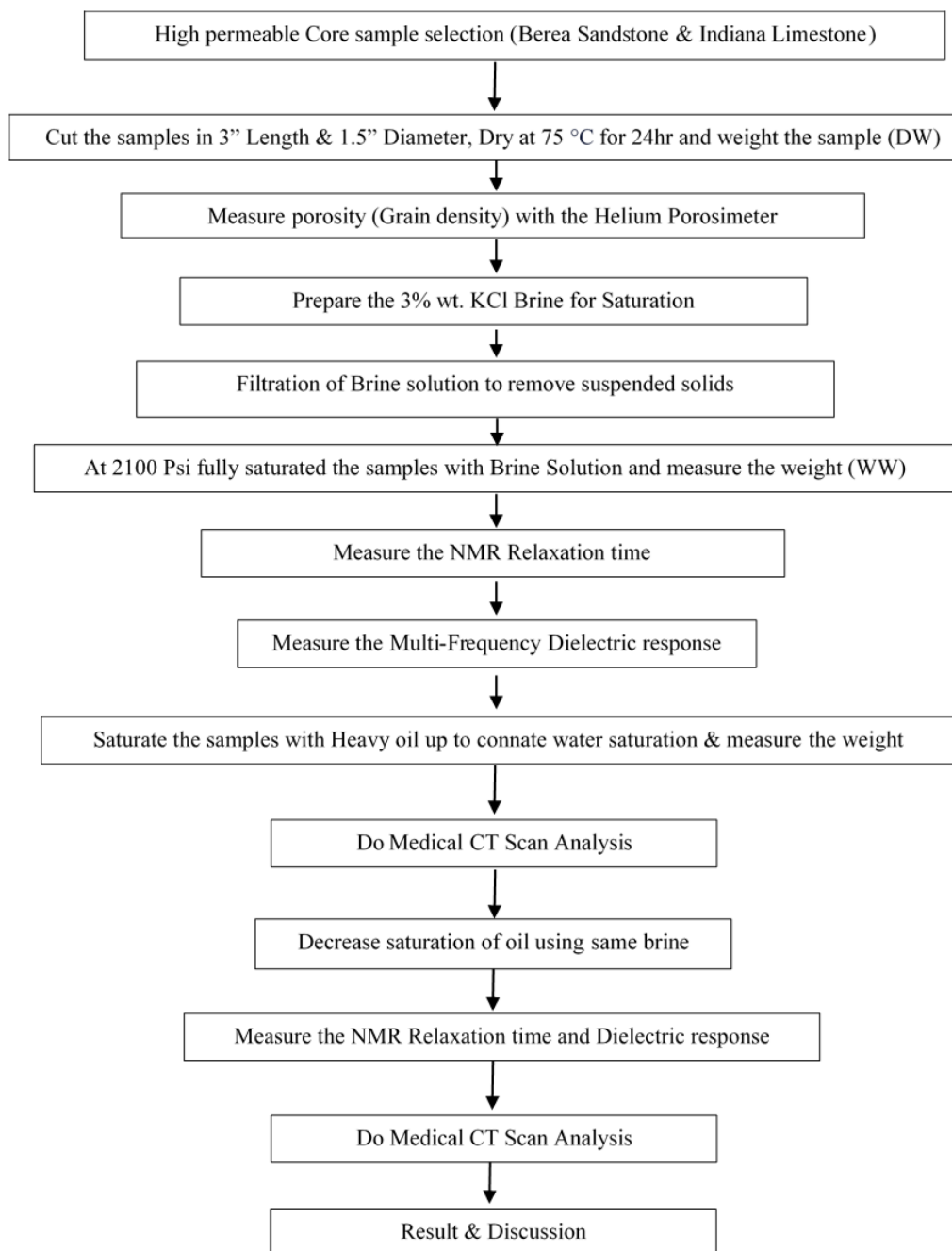


Figure 1—The process flow map of the experimental work for studying the porosity and oil saturation using NMR and dielectric methods.

Helium porosimeter was used to determine pore volume and porosity for the dried cores. The samples were placed in a helium porosimeter chamber, and the displacement of helium under controlled pressure changes was used to calculate the total pore volume. From the known bulk volume of each core plug, the absolute porosity was computed. These baseline porosity values formed the reference against which later porosity estimates (using NMR and dielectric methods) would be compared.

Brine Saturation and Baseline Measurement

Following porosity determination, the core plugs were fully saturated with a 3 wt.% potassium chloride (KCl) brine solution to achieve 100 percent water saturation ($S_{wi} = 100\%$). The brine was prepared by dissolving analytical-grade KCl in deionized water and filtering to remove any particulates. Each core was

first evacuated under vacuum for at least two hours to remove trapped air, then immersed in the brine. A pressure of 2,100 psi was applied to the brine-filled chamber to force the brine into all pore spaces. The cores remained under pressure for a minimum of 48 hours, to ensure full brine saturation. The saturated mass of each core was then measured, and saturation was confirmed gravimetrically by comparing the mass gain to the expected pore volume determined from the helium porosimeter.

With the cores at S_{wi} , baseline petrophysical measurements were taken. Nuclear magnetic resonance (NMR) measurement (Fig. 2a) was performed to capture the T_2 relaxation response of hydrogen nuclei in the pore fluids, providing an NMR-derived porosity estimate under fully water-saturated conditions. Multifrequency dielectric measurement (Fig. 2b) was also conducted, recording both the real and imaginary components of permittivity over a broad frequency spectrum to establish baseline conductivity and permittivity values that correspond exclusively to brine-filled pore spaces. These initial measurements provided a crucial reference for assessing the impact of heavy oil on subsequent NMR and dielectric responses.

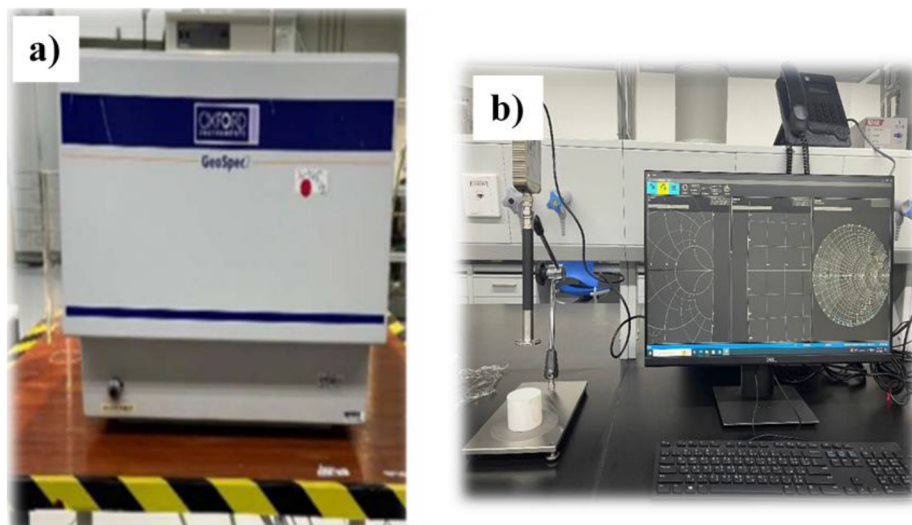


Figure 2—Laboratory setups used in this study, a) Nuclear Magnetic Resonance (NMR) Equipment b) Multifrequency Dielectric system

Heavy Oil Flooding and Irreducible Water Saturation

A heavy crude oil with an American Petroleum Institute (API) gravity of approximately 17° was selected to simulate heavy oil reservoir conditions. The oil's high viscosity at ambient temperature would normally impede displacement of brine, so its rheological properties were characterized across temperature to inform flooding conditions. Using a high-temperature rotational viscometer, the oil's dynamic viscosity was measured at three temperatures of 25 °C (4,652 cP), 85 °C (80.4 cP), and 110 °C (28.7 cP), the results are shown in Fig. 3. The marked decrease in viscosity with heating justified conducting core-flooding experiments at 110 °C to ensure effective oil mobility and realistic heavy oil displacement.

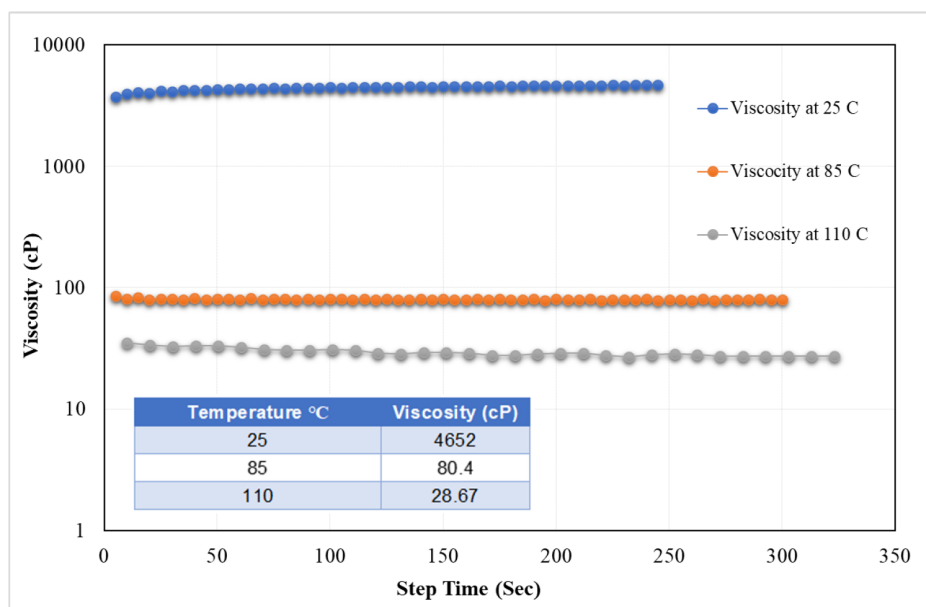


Figure 3—Effect of temperature (25 °C, 85 °C, and 110 °C) on the viscosity of heavy oil over time, illustrating substantial viscosity reduction at elevated temperatures.

Core flooding was conducted using a high-pressure and high-temperature (HPHT) apparatus comprising a core holder, syringe pumps, and a backpressure regulator. Each brine-saturated core plug was placed inside the core holder, then confining pressure was applied. Deionized water was pressurized around the sleeve to 1500 psi to simulate overburden stress and prevent fluid bypass. A backpressure of 500 psi was maintained at the core outlet for effective displacement of brine and ensure stable pore pressure. The entire assembly was heated to 85 °C, and once thermal equilibrium was reached (monitored via embedded thermocouples), heavy oil injection commenced. The experiment core flooding setup is shown in Fig. 4.

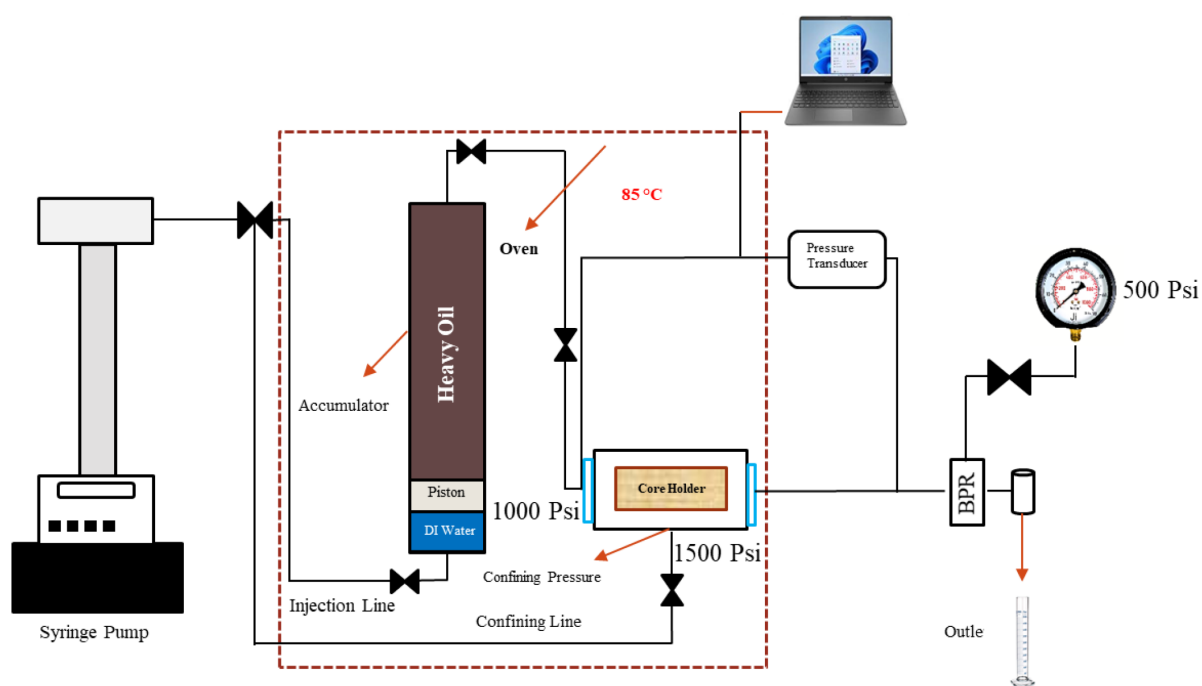


Figure 4—Schematic representation of the experimental core flooding setup used to saturate core samples with heavy oil at controlled pressure and temperature (85 °C), simulating reservoir conditions.

Heavy oil was injected at a constant flow rate of 0.1 cc/min using a precision syringe pump, delivering approximately three pore volumes of oil to displace brine. Throughout the flooding process, the produced brine was collected in graduated cylinders at the core outlet, and the cumulative brine volume was recorded periodically. Flooding continued until incremental oil injection yielded negligible additional brine production (<0.01 cc over one pore volume), indicating that irreducible water saturation (S_{wirr}) had been attained. The residual brine volume in each core was determined gravimetrically via production data, yielding S_{wirr} values of approximately 45% for Carbonate and 25% for the sandstone sample. Fig. 5 shows the heavy oil-saturated samples.

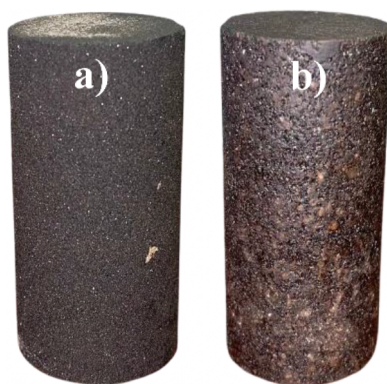


Figure 5—Core samples saturated with Heavy Oil a) Berea Sandstone b) Indiana Limestone

Intermediate Saturation Adjustment and Additional Measurement

To capture the evolution of NMR and dielectric responses as a function of fluid saturation, each core was subjected to one additional intermediate saturation stage beyond S_{wirr} . After initial S_{wirr} logging, a controlled volume of brine was reintroduced at different pressures, temperatures and injection conditions to adjust the oil saturation to an intermediate value (S_o), refer to Fig. 6. The cores were then allowed to equilibrate for at least two hours to minimize capillary and viscous transients.

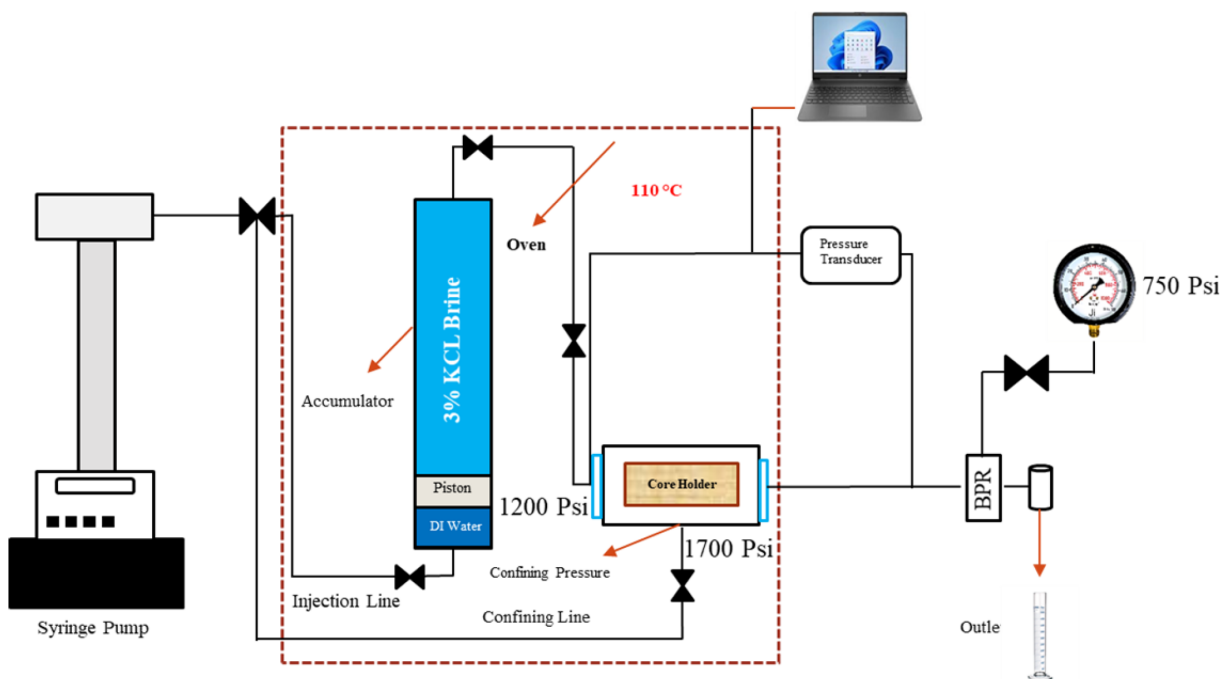


Figure 6—Schematic illustration of the core flooding setup utilized for desaturating core samples using 3% KCl brine at elevated temperature (110 °C)

At this intermediate saturation stage, both NMR and dielectric measurements were measured using the same procedure as at S_{wi} and S_{wirr} . NMR provided updated relaxation distributions, reflecting the new fluid mixture of heavy oil and residual brine, while dielectric measurement captured changes in permittivity and conductivity associated with the altered brine fraction. These intermediate readings, together with the baseline and S_{wirr} data, enabled the construction of saturation-dependent calibration curves for both NMR-derived porosity and dielectric permittivity.

CT Imaging and Comparative Analysis

To visualize the spatial distribution of fluids within the pore network, X-ray computed tomography (CT) scans were performed on each core plug at the conclusion of two saturation stages (S_{wirr} and S_o). Cores were sealed in plastic endcaps to prevent fluid exfiltration and scanned in a medical CT scanner at a voxel resolution of approximately 1 mm. The resulting image stacks were analyzed to see any heavy or viscous components entrapment along the core length.

Helium porosimeter established the true pore-volume benchmark, while porosity from NMR at full water saturation (S_{wi}) and value of saturation from dielectric measurements were collated and compared to this benchmark to assess each method's accuracy. At S_{wirr} and S_o , deviations of NMR-based porosity from helium porosimeter and gravimetric method revealed the extent to which heavy oil masked or altered the apparent pore volume. Similarly, saturation estimates from dielectric tools (based on permittivity–saturation relationships) were compared against gravimetric-derived S_{wirr} and S_o values to assess their reliability in heavy oil conditions.

The integrated methodology combining helium porosity, controlled brine and oil saturations, NMR and dielectric measurement at key saturation stages, and CT visualization can provide a robust framework for understanding and quantifying the effects of heavy oil on reservoir petrophysical measurements. By systematically varying saturation and validating results against conventional methods, this study demonstrates the enhanced accuracy of NMR and dielectric measurements for evaluating pore structures and fluid saturation, thereby significantly improving reservoir characterization for heavy oil recovery.

Results and Discussion

Baseline Porosity Characterization

Baseline porosity and permeability measurements were conducted using a helium porosimeter and gas permeameter, providing reference values independent of fluid influence. The porosity and permeability values obtained were 14.92% and ~200 mD for Indiana Limestone and 18.69% and ~200 mD for Berea Sandstone, establishing reliable benchmarks for further assessments, the results are listed in [Table 1](#).

Table 1—Porosity and permeability values for the core samples used in this study

Sample No	Diameter (mm)	Length (cm)	Pore Volume (cc)	Porosity (%)	Permeability (mD)
C1 (Carbonate)	37.9	7.452	12.54	14.92%	~200
S1 (Sandstone)	38.24	7.056	14.65	18.07%	~200

NMR Porosity Analysis

The NMR measurements for Indiana limestone reveal a pronounced dual-porosity system as shown in [Fig. 7](#), with both large interparticle voids and smaller intraparticle or vuggy pores contributing to the overall fluid storage. Under full brine saturation ($S_{wi} = 100\%$, 3 wt.% KCl at 2,100 psi, 48 h), the core exhibits a total porosity of 16.0 % and a bimodal T_2 distribution. A dominant peak near 631 ms representing brine in the larger pore bodies, and a secondary peak around 20–30 ms corresponding to water in the finer pores. When heavy oil is injected at irreducible water saturation (110 °C), the most viscous components preferentially lodge in the narrowest constrictions, collapsing the T_2 response into a single, short-relaxation peak at approx. 3.2 ms and reducing apparent porosity to 12.5 %. A subsequent warm-brine flood (85 °C, 500 psi backpressure) mobilizes only the lighter oil fractions from the larger pores, while the retained viscous residues remain bound within the finer network. This further diminishes the NMR-visible porosity to 7.9 % and effectively eliminates any long- T_2 signal. These results underscore how the hierarchical pore structure of Indiana limestone controls both fluid displacement and the evolving NMR signature, with dual-porosity behavior dictating enhanced trapping of viscous oil.

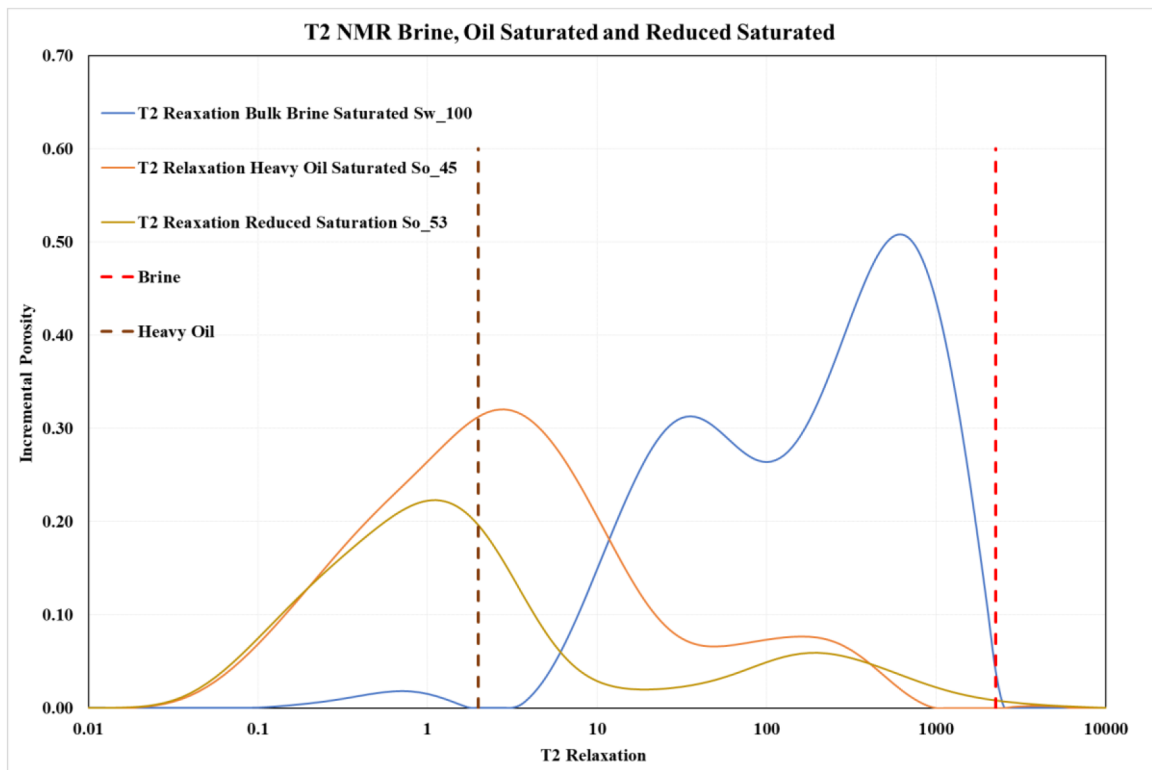


Figure 7—NMR response for Indiana limestone sample, at all the saturation stages.

By contrast, Berea sandstone behaves as a single-porosity medium as shown in Fig. 8, dominated by a uniform intergranular pore network. At full water saturation, the core displays a porosity of 19.3 % and a unimodal T_2 peak at around 177 ms, indicative of well-connected brine-filled pores. Heavy oil injection at irreducible water saturation collapses this response to a single, short-relaxation peak at approx. 2.5 ms, with measured porosity decreasing to 15.3 %. This shift reflects the irreversible entrapment of the oil's viscous components within the same pore space. A follow-on warm-brine flood displaces the more mobile oil fractions but fails to remove the most viscous residues from pore constrictions, driving porosity down to 9.8 % and shifting the residual T_2 signal further toward shorter relaxation times. The uniform pore architecture of Berea sandstone therefore produces a straightforward, single-peak NMR evolution, highlighting how single-porosity systems exhibit a simpler, but still substantial loss of mobile-fluid volume when challenged by heavy oil. Field NMR logs show the same patterns: vuggy carbonates commonly exhibit bimodal T_2 distributions, well sorted sandstones are typically unimodal, and in heavy oil intervals the viscous fractions push the signal to very short T_2 that the tool classifies as bound fluid, thereby lowering the apparent NMR porosity that is aligning with our lab based results (Morriss et al., 1994; Logan et al., 1998; Shao & Balliet, 2022).

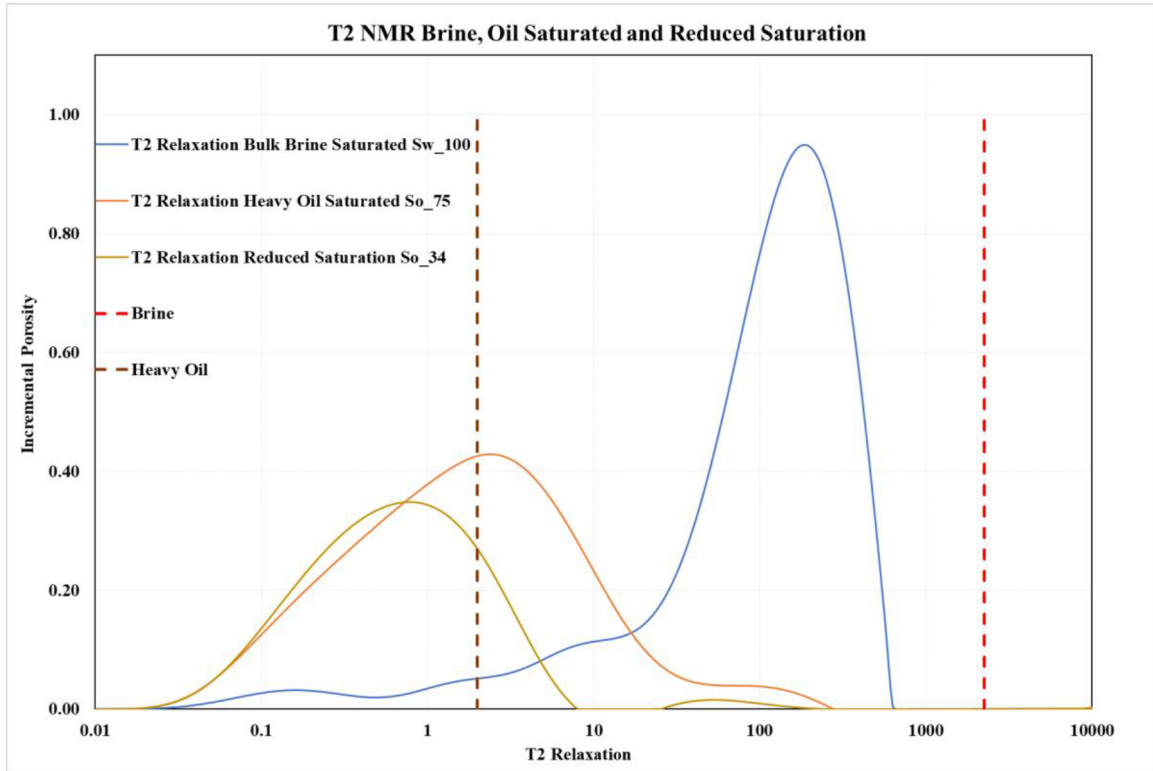


Figure 8—NMR responses for Berea sandstone sample, at all the saturation stages.

Dielectric Measurement and Saturation Analysis

Dielectric measurements were performed to analyze the permittivity and conductivity profiles of Indiana Limestone and Berea Sandstone samples at varying saturation conditions. Initially the equipment was calibrated using DI water having resistivity of 18.2 MΩ-cm which is similar to previous studies (Kesarwani et al., 2022; Khan et al., 2022). Dielectric permittivity and conductivity were measured at both the inlet and outlet points of the samples to assess fluid distribution and saturation homogeneity. Several models are available to evaluate and interpret these measurements; however, the Complex Refractive Index Model (CRIM) was chosen because of its higher accuracy and its capability to relate the measured permittivity directly to water saturation. CRIM relates the bulk dielectric permittivity to fluid saturations, expressed mathematically as:

$$\varepsilon_{CRIM} = \left((1 - \phi) \sqrt{\varepsilon_m} + \phi S_w \sqrt{\varepsilon_w} + (1 - S_w) \phi \sqrt{\varepsilon_{HC}} \right)^2$$

ε_{CRIM} is the bulk measured dielectric permittivity, where ϕ represents the porosity derived from the helium porosimeter, ε_m is the matrix permittivity based on volumetric mineralogy, ε_w denotes the permittivity of brine influenced by salinity and temperature, ε_{HC} indicates the hydrocarbon permittivity (gas = 1, oil = 2), and S_w is the water saturation.

Dielectric responses for Indiana Limestone and Berea Sandstone at different saturation levels, measured at both the inlet and outlet, are shown in Fig.9 and Fig. 10. Analysis of permittivity profiles indicates clear distinctions among different saturation conditions, particularly at lower frequencies where saturation sensitivity is heightened. Conductivity profiles similarly reflect fluid saturation variations, with noticeable differences at higher frequencies, corresponding to fluid distribution dynamics. By applying the CRIM model to these measured dielectric parameters, accurate estimations of water-filled porosity were achieved, highlighting the model's effectiveness in characterizing saturation-dependent dielectric properties.

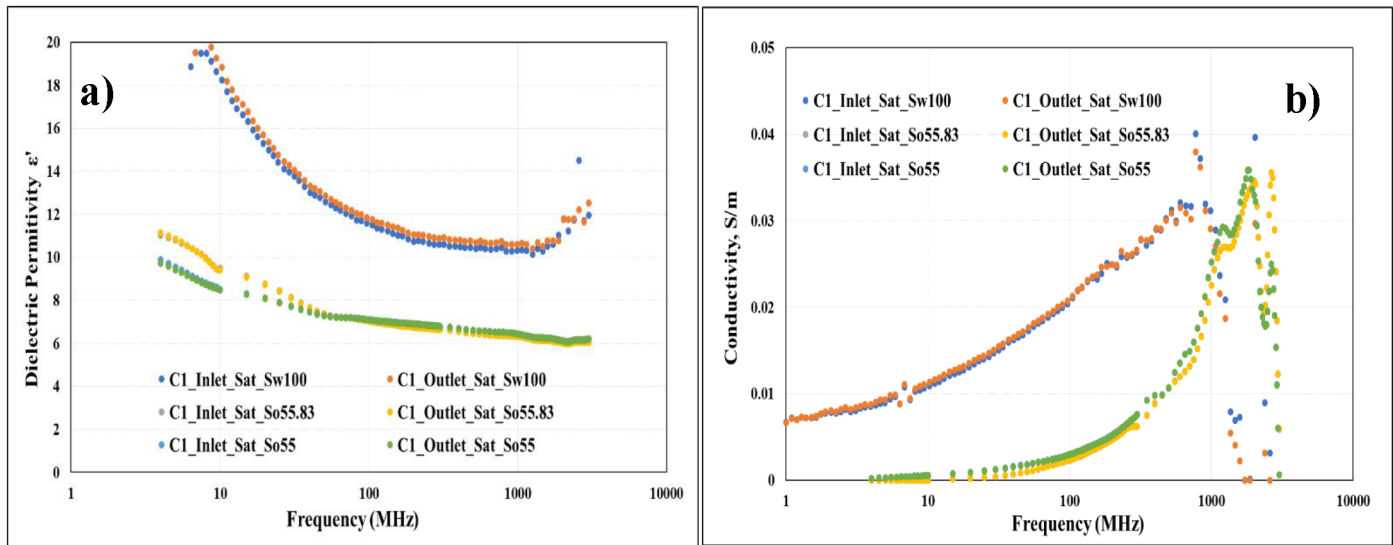


Figure 9—The multifrequency dielectric results (real permittivity (a) and conductivity (b)) of Indiana Limestone at different saturation stages.

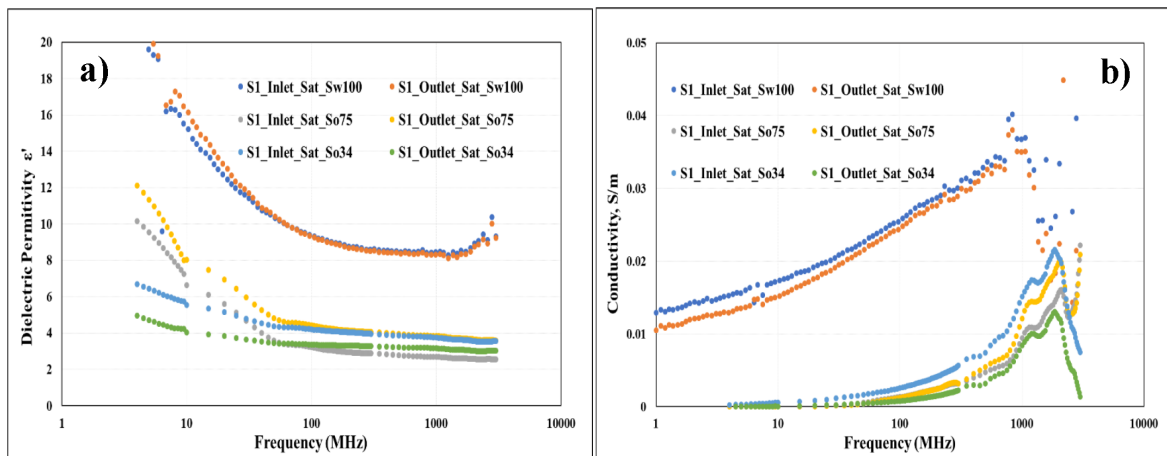


Figure 10—The multifrequency dielectric results (real permittivity (a) and conductivity (b)) of Berea Sandstone at different saturation stages.

In this study, matrix permittivity (ϵ_m) of 4.5 was used for the limestone sample while a value of 3.0 was used for sandstone. A hydrocarbon permittivity (ϵ_{HC}) of 2 was used for the heavy oil, while the brine permittivity (ϵ_w) was measured at 62. The helium porosities from Table 1 (14.92% for Indiana limestone, 18.07% for Berea sandstone) were used in the CRIM model. The measured bulk permittivity ϵ_{CRIM} at each saturation state was used to compute water saturation (S_w) via the CRIM mixing law. The resulting saturations (listed in Table 2 for limestone, and Table 3 for sandstone) agree closely with the gravimetrically measured values at $S_w = 100\%$ and S_{wir} . For example, Indiana Limestone at full water saturation ($S_w = 100\%$) had a CRIM permittivity value (ϵ_{CRIM}) of 10.14 and a CRIM-calculated S_w of 100%, matching the measured saturation. At irreducible water saturation (S_{wir}), ϵ_{CRIM} decreased to 6.19, yielding a calculated S_w of 48.15% compared to the measured 45%. During the oil saturation stage (S_o), ϵ_{CRIM} was 6.30, corresponding to a calculated S_w of 51.26% versus a measured S_w of 47% and oil saturation of 53%. In Berea Sandstone, ϵ_{CRIM} decreased from 8.20 at full water saturation ($S_w = 100\%$, calculated $S_w = 100\%$) to 3.76 at S_{wir} , giving a calculated S_w of 22.66% compared to the measured 25%. At the oil saturation stage (S_o), ϵ_{CRIM} was further reduced to 3.64, with calculated S_w of 20%, significantly lower than the measured S_w of 64% ($S_o = 34\%$). This discrepancy suggests possible underestimation of residual water by CRIM because

of disconnected water phases as the dielectric current only passes through connected water saturations. Overall, as brine was progressively replaced by non-conductive heavy oil, both permittivity and conductivity significantly decreased. The effective permittivity dropped approximately by half in both samples from full water saturation to S_{wirr} (limestone: ϵ reduced from 10.1 to 6.2; sandstone: ϵ reduced from 8.2 to 3.8), reflecting the replacement of high-permittivity brine. Bulk conductivity similarly showed a dramatic decline due to the reduction in pore fluids. Field multifrequency dielectric logs show analogous behavior, with formation permittivity and conductivity decreasing as brine is replaced by hydrocarbons and with CRIM style mixing giving consistent S_w when water is well connected, while disconnected phases or invasion can bias the interpretation, which is consistent with our observations (Freed et al., 2018; Han & Misra, 2018; Wang et al., 2018).

Table 2—Saturation from dielectric and flooding experiments for Indiana limestone.

Dielectric Permittivity (ϵ_{CRIM})	Porosity (%)	Saturation (%)	Saturation based on flooding experiment	Accuracy
10.14	14.92	Brine Saturation at $S_w = 100\%$	$S_w = 100\%$	100%
6.19	14.92	At $S_{wirr} = 48.15\%$	$S_{wirr} = 45\%$	93.45%
6.30	14.92	At $S_w = 51.26\%$, $S_o = 48.74\%$	$S_w = 47\%$, $S_o = 53\%$	91.68%

Table 3—Saturation from Dielectric Measurement of Berea sandstone

Dielectric Permittivity (ϵ_{CRIM})	Porosity (%)	Saturation (%)	Saturation based on flooding experiment	Accuracy
8.2	18.07	Brine Saturation at $S_w = 100\%$	$S_w = 100\%$	100%
3.76	18.07	At $S_{wirr} = 22.66\%$	$S_{wirr} = 25\%$	90.64%
3.64	18.07	At $S_w = 20\%$ $S_o = 80\%$	$S_w = 64\%$, $S_o = 34\%$	31.25%

CT Scan Visualization

At irreducible water saturation, the Indiana limestone's pore network is saturated with heavy oil, appearing as deep orange–red with brighter patches indicating higher oil concentration as shown in Fig. 11a. After flooding at elevated temperature with the same 3wt.% KCl brine, the lighter oil fractions are swept away, leaving behind the densest heavy-oil components shown in Fig. 11b. These heavy residues occupy pore spaces previously filled by lighter oil, reducing apparent porosity and shifting overall CT density in the pores toward red, evidence of the remaining high-density oil. The Berea sandstone exhibits similar but more pronounced behavior. At irreducible water saturation, the core body appears predominantly dark red, indicating significant heavy-oil saturation as per Fig. 12a. After flooding with hot 3wt.% KCl brine, nearly all lighter oil fractions are removed, leaving only the densest heavy oil trapped within the core body shown in Fig. 12b. Consequently, fewer open pores remain (lower apparent porosity). Additionally, brine becomes increasingly disconnected within the core because of this trapped heavy oil component. This disconnected brine phase explains the discrepancies observed in earlier dielectric measurements for the Berea sandstone.

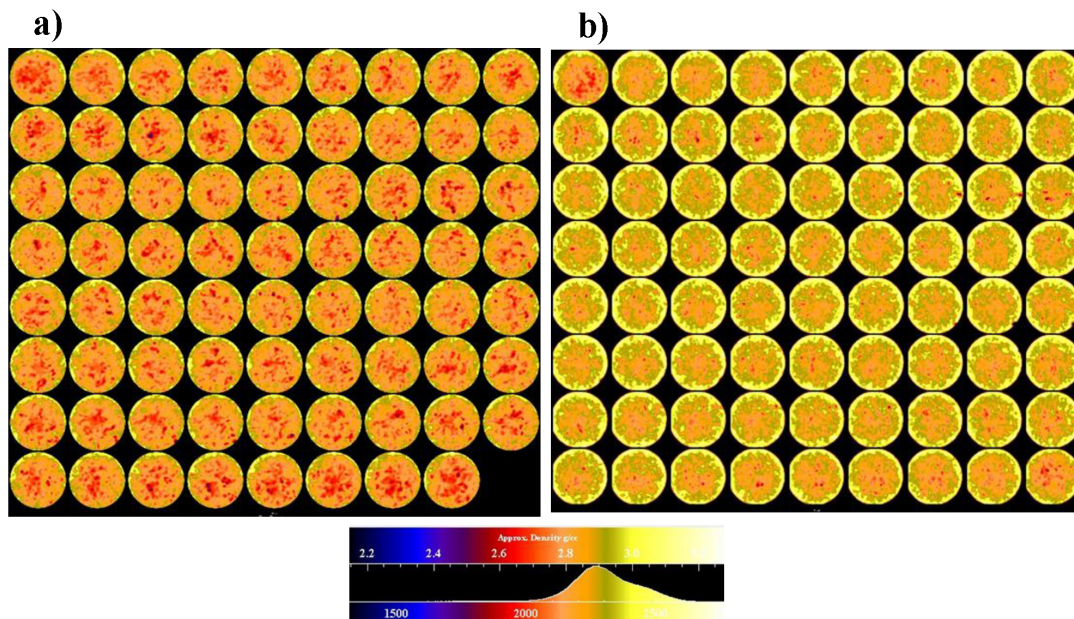


Figure 11—CT image scan of Indiana Limestone at a) S_{wirr} and b) at S_o .

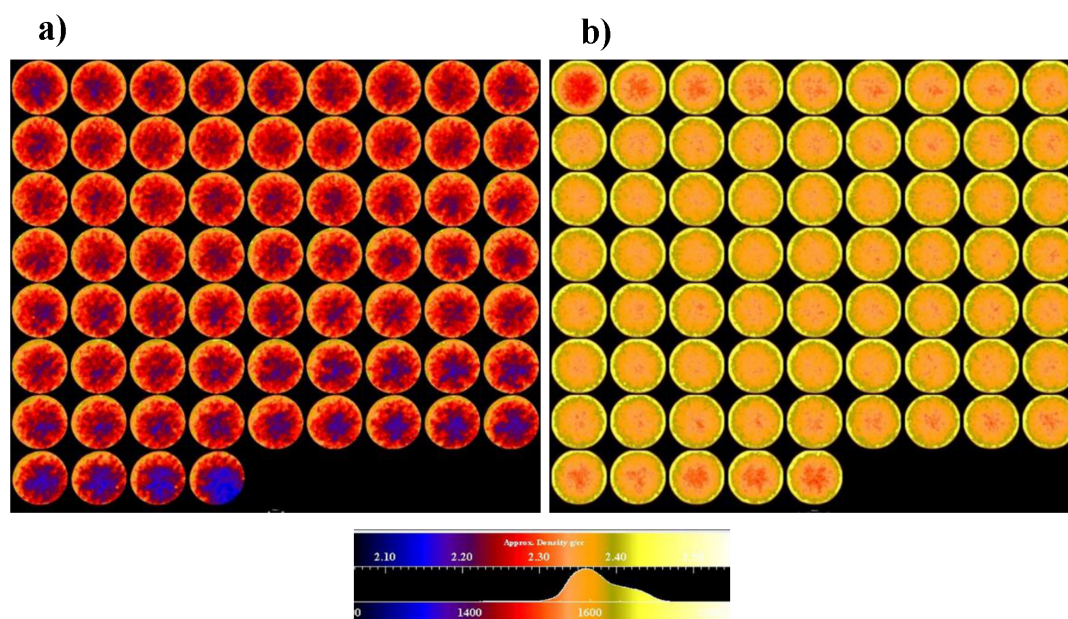


Figure 12—CT image scan of Berea Sandstone at a) S_{wirr} and b) at S_o .

Method Comparison and Validation

When we look at porosity, the gravimetric (weight-based) method (presented in Table 4) and NMR tell very different stories once heavy oil and brine floods enter the picture. Gravimetrically, the Indiana limestone core shows porosities of 14.54 % at full-water saturation, 15.53 % at irreducible water (S_{wirr}), and 15.38 % after the reduced saturation (S_o), values nearly identical to the 14.92 % baseline from helium porosimeter. In contrast, NMR porosity for that same core starts at 16.0 % with full brine, falls to 12.5 % at S_{wirr} , and plunges to just 7.9 % after flood, revealing that over half of the pore volume is occupied by trapped and viscous oil that gravimetry simply counts as filled. The Berea sandstone shows the same divergence where the gravimetric porosities of 18.09 %, 19.84 %, and 19.60 % for the full-water, S_{wirr} and S_o , respectively, which somehow remain close to the 18.07 % obtained from helium porosimeter. However, NMR porosity drops from 19.3 % (full-water) to 15.3 % (S_{wirr}) and down to 9.8 % (S_o), again exposing the large volume of heavy-oil residue invisible to weight methods.

Table 4—Porosities for carbonate and sandstone samples, measured through the gravimetric methods.

Samples	Saturation	Dry Weight (gm)	Wet Weight (gm)	Effective Density of Fluid (g/cm ³)	Bulk Volume (cm ³)	Porosity (%)
Indiana Limestone	Sw = 100%	189.08	201.54	1.02	84.07	14.54
	Swirr = 45%	189.08	201.54	0.954	84.07	15.53
	So = 53%	189.08	201.54	0.9636	84.07	15.38
Berea Sandstone	Sw = 100%	174.57	189.52	1.02	81.04	18.09
	Swirr = 25%	174.57	189.52	0.93	81.04	19.84
	So = 34%	174.57	189.52	0.9408	81.04	19.60

When benchmarking dielectric-CRIM against gravimetric saturations as shown in Table 2 and 3, both rocks show excellent agreement under full-water and irreducible-water conditions, but diverge after the warm-brine flood in the Berea sandstone only. In the Indiana limestone, CRIM predicts $S_w = 100\%$ exactly at full-water saturation (100% accuracy) and remains reliable at S_{wirr} , calculating 48.15% compared to the measured 45%, resulting in a small error of 3.15 percentage points (~94% accuracy). After the high-temperature 3% KCl flood, CRIM continues to provide reliable water saturation estimates (51.26% compared to the gravimetric value of 47%), yielding a modest error of 4.26 percentage points (~92% accuracy).

The Berea sandstone shows similar accuracy at initial conditions: CRIM matches perfectly at full-water saturation (100%, 100% accuracy) and closely at S_{wirr} (22.66% versus measured 25%, error of 2.66 percentage points, ~91% accuracy). However, after warm-brine injection, CRIM's accuracy diminishes significantly in the Berea sandstone, reporting only 20% water saturation compared to the gravimetrically measured 64%, resulting in an error of 44 percentage points (~32% accuracy). This substantial decrease in accuracy is because of heavy oil invasion in Berea sandstone clearly leads to loss of water-phase connectivity, isolating brine in pockets that no longer contribute to the bulk dielectric response and causing simple models like CRIM to underestimate true water saturation. Mechanistically, this discrepancy arises not from significant wettability alteration (which is unlikely in a water-wet sandstone, unlike the pronounced wettability changes often seen in carbonates) but from phase redistribution: heavy oil becomes the dominant continuous phase occupying pore throats and breaking apart the formerly connected water pathways. As a result, water is stranded in disconnected films or droplets, severing the continuous electrical network. To correct such interpretation errors, more advanced dielectric mixing models that account for partial connectivity or percolation effects can be applied. For example, the Hanai–Bruggeman effective medium model or the Licht Necker–Rother logarithmic mixing rule introduce geometric/exponential weighting factors that reduce the contribution of the high-permittivity brine when it is not fully connected, leading to more accurate saturation estimates. Furthermore, a joint inversion of dielectric and NMR measurements

offers a robust solution: NMR captures the total fluid-filled porosity and distinguishes oil vs. water (including heavy oil and any isolated water) by their hydrogen content, while dielectric measurements specifically reflect the volume of electrically connected water. By integrating these datasets, one can identify the fraction of water that is electrically disconnected (seen by NMR but invisible to dielectric), thereby yielding more reliable true water saturation values in heavy oil-rich formations.

Conclusion

This investigation confirms that Nuclear Magnetic Resonance (NMR) and multifrequency dielectric-dispersion techniques can accurately determine porosity and hydrocarbon saturation in heavy-oil reservoirs without external calibration, with an exception noted in Berea sandstone under certain conditions. The reference pore volume was provided by helium porosimeter, and both NMR porosity and dielectric saturation measurements closely matched this reference value at complete brine saturation. Heavy oil saturation caused distinct changes in petrophysical parameters, notably observed in NMR porosity results, which showed significant reductions, decreasing from 19.3% to 9.8% in sandstone (a 49% reduction) and from 16% to 7.9% in carbonate (approx. 51% reduction). These changes were supported by decreased amplitudes and changes in T_2 relaxation and visually confirmed by CT imaging, indicating heavy oil presence within the pores. Dielectric dispersion provided consistent and reliable saturation results across various saturation levels, consistently recording reductions in permittivity and conductivity as brine was replaced by heavy oil and vice versa. Dielectric saturation values matched gravimetric saturation measurements with high accuracy, demonstrating the method's reliability during the drainage and imbibition process, except for the Berea sandstone where significant deviations occurred post warm-brine flooding. The investigation identified that sandstone formations, particularly Berea sandstone, present more significant challenges compared to carbonate formations for dielectric dispersion saturation calculation. In Berea sandstone, dielectric methods showed a substantial accuracy decrease after warm brine flooding (20% versus the gravimetric measured 64% water saturation, an error of 44 percentage points, ~32% accuracy), whereas carbonate samples maintained higher accuracy (~92%). Key limitations include challenges in accurately interpreting dielectric responses under significant fluid redistribution conditions and the complexities introduced by heavy oil trapping, especially in sandstone reservoirs.

In heavy-oil reservoirs, NMR logging should be prioritized whenever conventional neutron or density porosity logs might misidentify heavy oil as formation water due to heavy oil's high hydrogen content and near water-like density. NMR directly measures the liquid-filled pore volume and differentiates fluids by their characteristic relaxation signatures, a critical advantage in viscous oil zones where heavy oil yields distinctive NMR responses. Use dielectric to estimate S_w where brine is expected to be well-connected (e.g., stable salinity, minimal invasion). If heavy oil disrupts water connectivity (e.g., post-flood Berea sandstones), dielectric can under-estimate S_w , cross-check with NMR before final interpretation. Integrating NMR, dielectric dispersion, and CT imaging into a comprehensive workflow significantly enhances the accuracy of porosity and saturation measurements. This approach supports more reliable reservoir characterization, facilitates optimized recovery strategies, and ultimately enhances heavy oil production efficiency.

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